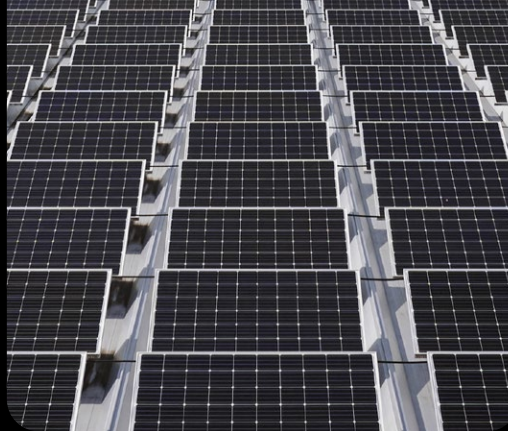




Environmental Progress Report



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Reflections

At Apple, we are constantly innovating to make the world’s best technology while reducing our impact on the environment. That means taking urgent steps to fight climate change and working collaboratively with our suppliers and local communities.

Every year, teams across our company find new and innovative ways to make our technology better for people and the planet. That progress has brought us closer than ever to Apple 2030: our goal to become carbon neutral for our global footprint, including our supply chain and the energy our customers use to power their devices.

Thanks to these efforts, I’m proud to share that Apple has now cut our overall emissions by more than 60 percent since 2015. This report covers the work that makes a milestone like that possible. It describes in detail how we continue to dramatically reduce our climate impact, while helping accelerate a global transition toward recycled materials and renewable energy.

Since 2018, we’ve powered every Apple facility with renewable energy — including our offices, retail stores, and data centers. That progress is quickly making its way across our global supply chain, and today, our suppliers now support more than 17.8 gigawatts of clean energy around the world.

But our commitment doesn’t end there. We’re also investing in clean energy projects to match the energy our customers use to charge their devices. And with our Power for Impact program, we’ve launched renewable energy projects in countries including the Philippines, Thailand, and South Africa. By expanding access to safe, reliable electricity, we can protect the planet and support the communities most significantly impacted by climate change.

To drive down carbon emissions even further, we’re using more recycled materials than ever before. We now use 99 percent recycled rare earth elements in the magnets across our products. We’ve also continued to scale the use of recycled materials like tungsten, aluminum, cobalt, gold, and lithium. And we’ve done all that while making our products even more durable and easy to repair, because technology that lasts longer is better for the environment and our customers’ wallets.

It is amazing to see this progress come together in a single product. In October, we unveiled our latest carbon neutral product, Mac mini, which is made with more than 50 percent recycled content and manufactured with electricity exclusively sourced from renewable energy. That’s helped reduce the carbon footprint of Mac mini by 80 percent from our business-as-usual scenario. And to balance the remaining emissions that can’t be avoided, we invest in high-quality, nature-based projects that reduce carbon in the atmosphere.

These projects are helping restore ecosystems around the world. Last year, I visited an area of the Atlantic Forest in Brazil that was fully deforested not long ago. Thanks to the Restore Fund we launched with our partners, it’s now a flourishing working forest — filled with native tree species that could have been lost forever. It was a powerful reminder that with dedication and effort, we can bring new life to once-thriving habitats.

At Apple, we’ve always believed that innovation is a powerful force for progress. So we’ll continue to do our part to protect the planet — and to build a better future for generations to come.

Lisa Jackson
VP, Environment, Policy and Social Initiatives



Report highlights

Reduced our overall GHG emissions by more than 60 percent

We reduced our gross greenhouse gas (GHG) emissions across scopes 1, 2, and 3 by more than 60 percent compared with our 2015 baseline year — not including offsets.¹ In that same time period, revenue grew by more than 65 percent. We estimate that we’ve avoided 41 million metric tons of emissions this year through reduction efforts like transitioning our supply chain to renewable electricity and sourcing recycled content.

➔ [Read more on page 12.](#)



Increased our use of recycled materials

We’re making steady progress on our journey toward using only recycled or renewable materials in our products. In 2024, approximately 99 percent of tungsten, 71 percent of aluminum, 53 percent of lithium, and 40 percent of gold; and 76 percent of cobalt in our products came from recycled sources.³ In 2024, 24 percent of the materials we shipped in Apple products, by weight, came from recycled sources.⁴

➔ [Read more on page 15.](#)

Enhanced repairability features

In 2024, we introduced design and software features to improve device repairability. The iPhone 16 debuted a new, faster process for removing the battery from the enclosure — using low-voltage electricity. We’ve improved support for third-party parts used in repair. And Repair Assistant, launched with iOS 18, allows customers and repair professionals to configure new and used Apple parts directly on the device.

➔ [Read more on page 40.](#)



Launched the first carbon neutral Mac

In October 2024, we announced the carbon neutral Mac mini, made with over 50 percent recycled content. The electricity used to manufacture Mac mini is sourced from 100 percent renewable energy. We’ve also invested in clean energy projects worldwide to address product use and prioritized lower-carbon shipping modes to reduce emissions. These actions have reduced the carbon footprint of Mac mini by over 80 percent from a business-as-usual scenario.² To offset the remaining emissions, we apply high-quality carbon credits from nature-based projects.

➔ [Read more on page 20.](#)

Expanded the use of recycled rare earth elements across our products

We now use 99 percent recycled rare earth elements in all magnets of our products. In 2024, more than 80 percent of the total rare earth elements that we shipped in products came from certified recycled sources — up from 75 percent in 2023.

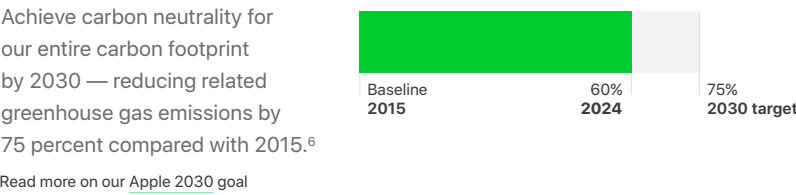
➔ [Read more on page 15.](#)



Goals and progress

Emissions

Apple 2030 is our science-based commitment to achieving carbon neutrality for our entire carbon footprint, including transitioning our entire value chain to 100 percent clean electricity.⁵



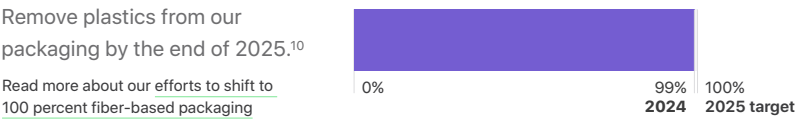
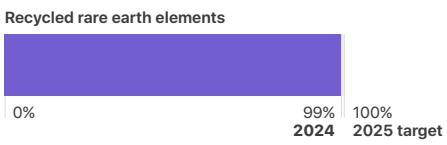
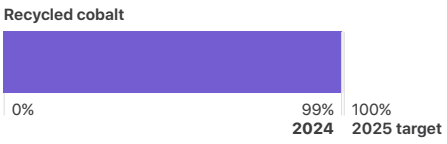
Materials

We’re committed to one day using only recycled or renewable materials in our products and packaging, and to enhance material recovery.⁷

Transition to 100 percent recycled cobalt, tin, gold, and rare earth elements in select components and applications by the end of 2025.⁸

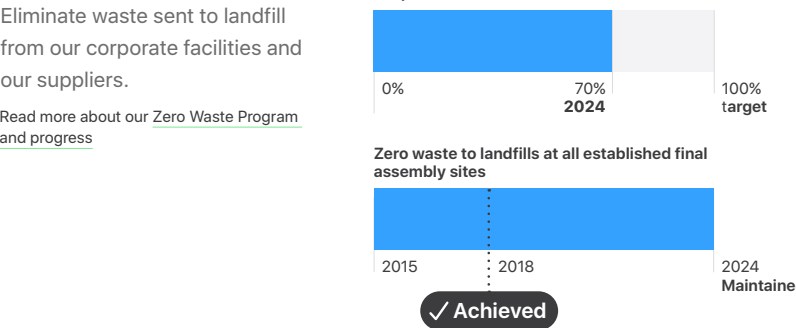
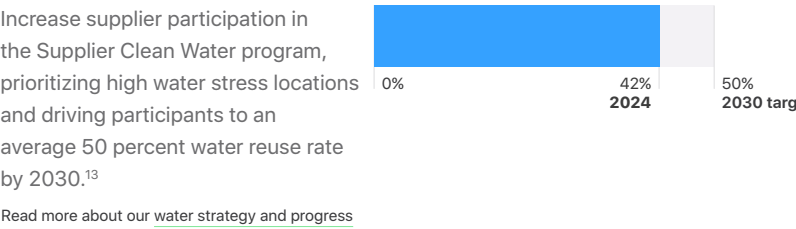
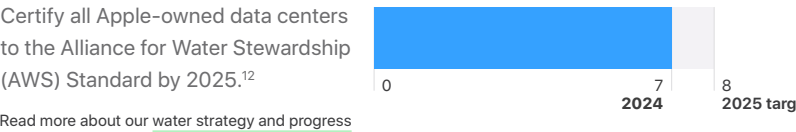
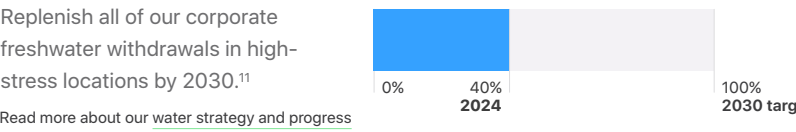
- Gold plating in all Apple-designed rigid and flexible printed circuit boards
- Tin soldering in all Apple-designed rigid and flexible printed circuit boards
- Cobalt in all Apple-designed batteries⁹
- Rare earth elements in all magnets

Read more on the [expansion of recycled materials in our products](#)



Resources

We’re committed to stewarding water resources and working to eliminate waste sent to landfills.



Environmental Initiatives

In this section

Overview

Apple 2030

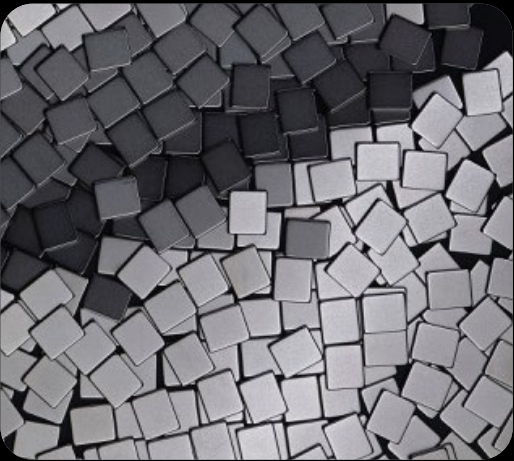
- Apple 2030 journey
- Approach
- Design and materials
- Electricity
- Direct GHG emissions
- Carbon removal

Resources

- Approach
- Product longevity
- Material recovery
- Water
- Zero waste

Smarter Chemistry

- Approach
- Mapping
- Assessment
- Innovation



Overview

Environmental Initiatives

Apple 2030

Apple 2030 is our commitment to be carbon neutral for our entire carbon footprint. Our journey to 2030 is focused on first reducing our scope 1, 2, and 3 greenhouse gas emissions by 75 percent compared with 2015, and investing in high-quality carbon removal solutions for the remaining emissions.

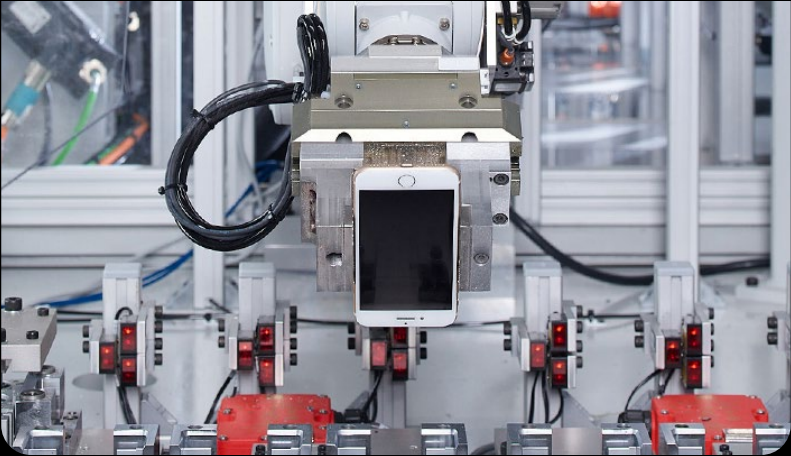
- Design and materials
- Electricity¹⁴
- Direct GHG emissions
- Carbon removal



Resources

We aim to make durable, long-lasting products and enhance material recovery. And we're committed to stewarding water resources and working to eliminate waste sent to landfills.

- Product longevity
- Material recovery
- Water
- Zero waste



Smarter Chemistry

Through innovation and material selection, we design our products to be safer for anyone who assembles, uses, or recycles them — and to have less of an impact on the environment.

- Mapping
- Assessment
- Innovation



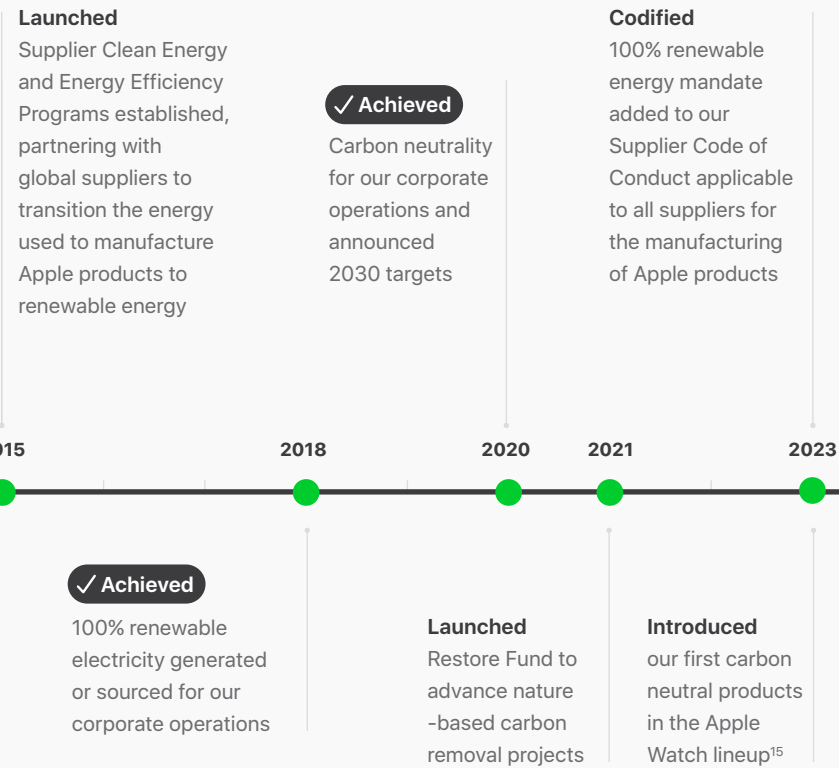
Apple 2030

- In this section**
- Apple 2030 journey
 - Approach
 - Design and materials
 - Electricity
 - Direct GHG emissions
 - Carbon removal



Apple 2030 journey

Where we've been



2024 activities



Where we're going



Approach

Apple 2030

We’re committed to our ambitious, science-based Apple 2030 goal to reduce our collective scope 1, 2, and 3 emissions — upstream and downstream — by 75 percent before balancing the remaining emissions with high-quality carbon removals. To achieve this goal, we’re reducing emissions across our value chain and directing our efforts toward decarbonizing the largest sources of emissions. We’ve cut emissions across our value chain by more than 60 percent since 2015. During the same period, our revenue grew by more than 65 percent.

We accelerated our progress with our transition to sourcing 100 percent renewable electricity for our offices, retail stores, and data centers in 2018. And in 2020, we achieved carbon neutrality for our corporate emissions, which we’ve maintained annually.²²

Decarbonizing our supply chain is a crucial component of our efforts. We’re driving progress by expanding our suppliers’ sourcing of renewable energy and increasing the use of recycled and renewable materials in manufacturing our products. Our initiatives emphasize achieving emissions reductions before applying high-quality, nature-based credits.

Our goal is consistent with the Intergovernmental Panel on Climate Change’s (IPCC) recommendation for global carbon neutrality.²³ We’re also committed to working toward reaching a 90 percent reduction in emissions from our 2015 baseline by 2050. Achieving deep decarbonization will require a collective, global effort across industries and economies. And while reaching a 90 percent reduction in emissions is outside Apple’s or any one company’s control, we’re committed to taking actions that support this goal. To catalyze change beyond our footprint, we’re engaging with stakeholders and communities to identify opportunities to expand the use of recycled materials and renewable energy. For more details, read our [Engagement and Advocacy](#) section.

Apple 2030 roadmap



Design and materials
Designing products and manufacturing processes to be less carbon intensive through thoughtful material selection, increased material efficiency, greater product energy efficiency, the use of recycled and renewable materials in our products and packaging, and enhanced material recovery
[Read more on page 13](#)



Electricity
Increasing energy efficiency at our facilities and transitioning the electricity throughout our entire value chain — including manufacturing and our customers’ product use — to 100 percent clean electricity by 2030
[Read more on page 23](#)



Direct GHG emissions
Reducing direct greenhouse gas emissions across our facilities and supply chain through process innovation, emissions abatement, and shifting away from fossil fuels
[Read more on page 31](#)



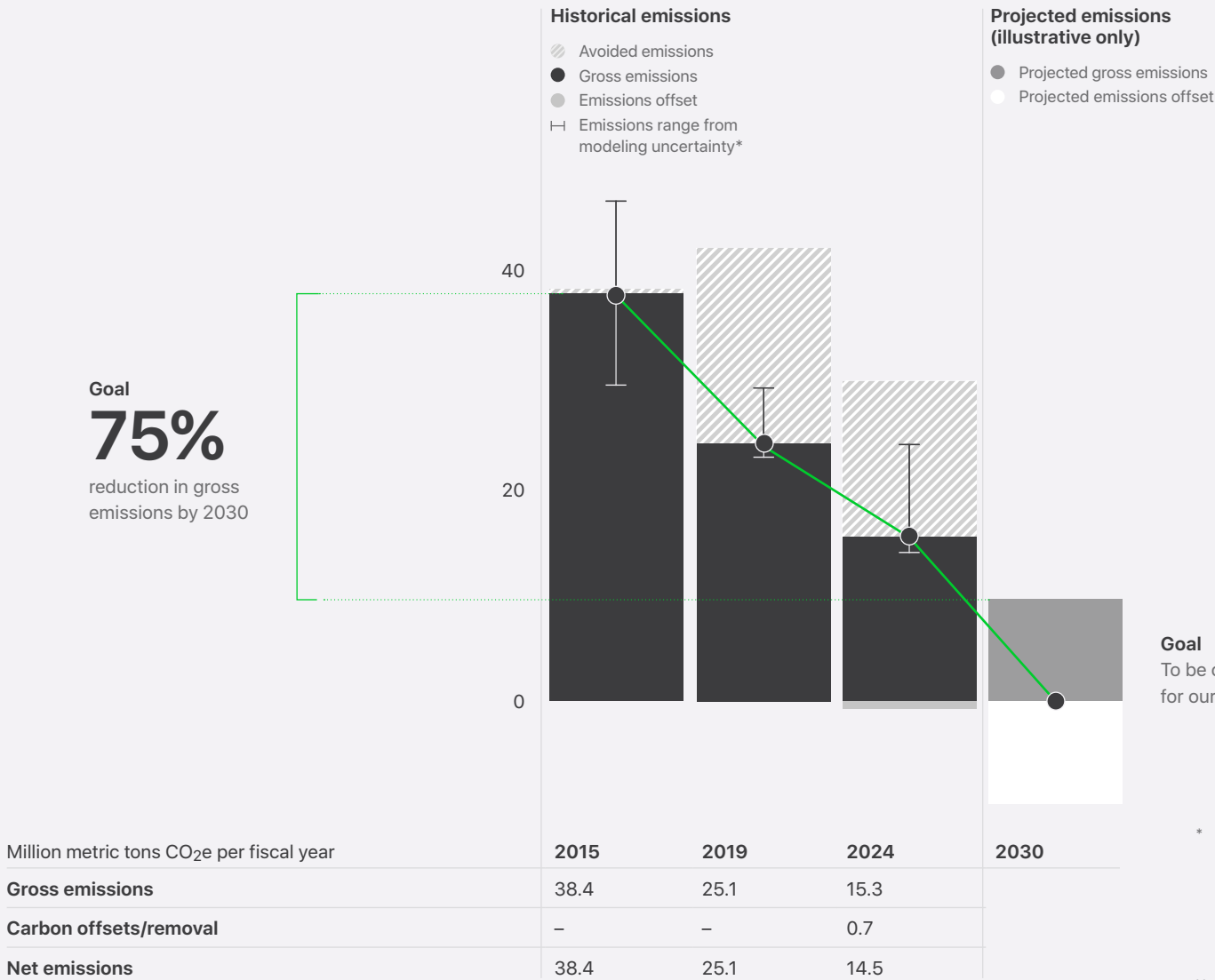
Carbon removal
In parallel with our emissions reduction efforts, scaling up investments in carbon removal projects, including nature-based solutions that can protect and restore ecosystems worldwide
[Read more on page 35](#)

≥75%
emissions reduction

≤25%
remaining footprint

Apple’s progress toward carbon neutrality

We’ve reduced our entire carbon footprint by more than 60 percent compared with 2015



Achieving progress that benefits business

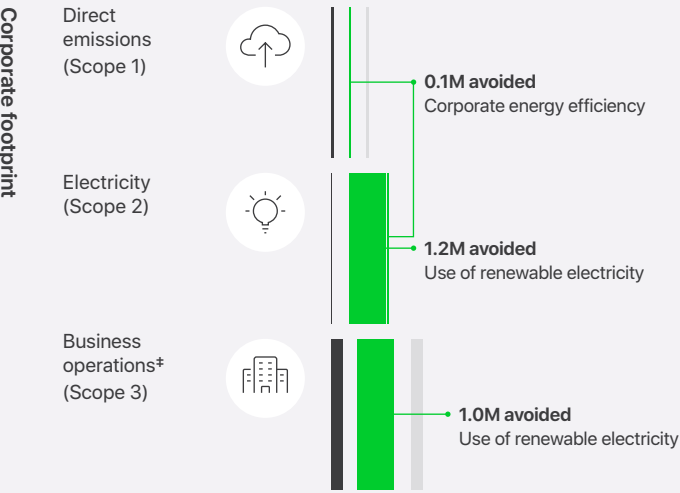
We underpin our climate strategy with strong business principles and innovation, harnessing the power of markets to replicate our solutions at scale — aiming to create the impact necessary to meet global reduction targets. We’re also working to accelerate the global transition toward decarbonization, exploring the next generation of solutions, and advocating for more effective policies. These efforts include fostering new and strengthened collaborations in public and private partnerships and investing in technological advancements. We’ve issued \$4.7 billion in green bonds to model how businesses can drive investments to reduce global emissions. Our most recent offering in 2019 supported 74 global projects that will mitigate or offset 33.4 million metric tons of CO₂e. To learn more about our work, read our [Green Bond Impact Report](#).

Science and transparency guide our efforts. Our emissions reduction targets align with what current climate science shows is necessary to adhere to a 1.5° C pathway with no or limited overshoot. We calculate our footprint across our entire value chain, covering both direct and product-related emissions — from sourcing materials to end of life. We use the results of our detailed carbon accounting to adjust our Apple 2030 roadmap, which lays out our plan to become carbon neutral. And we’re committed to disclosing our carbon footprint, as well as our climate strategy and progress. Transparency and disclosure are essential not only for sharing our climate strategy and progress, but also for inviting others to collaborate with us. Our annual Environmental Progress Report, as well as our response to the global disclosure nonprofit CDP, provides details on our progress. Read our latest response to the [CDP Climate Change 2024 questionnaire \(PDF\)](#).

And progress needs to include low-income and historically marginalized communities that often bear the brunt of climate change. We’re pursuing ways to directly support these communities in our climate programs. To learn more about our work, read our feature on our [Power for Impact program](#).

Apple’s comprehensive carbon footprint

Metric tons CO₂e



● Gross emissions ● Avoided emissions ● Emissions offset

Total avoided emissions

41Mmt

Net Greenhouse Gas Emissions*

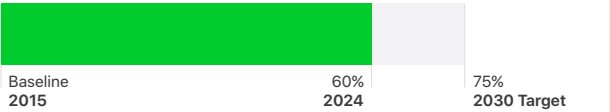
14.5Mmt

Goal

Achieve carbon neutrality for our entire carbon footprint by 2030 — reducing related emissions by 75 percent compared with 2015.

Progress

We estimate that in 2024, our environmental programs avoided 41 million metric tons of emissions across all scopes. Our long-standing initiatives continue to yield clear results, including sourcing 100 percent renewable energy for our facilities, transitioning suppliers to renewable energy, and using low-carbon materials in our products.²⁴ While our revenue has grown by more than 65 percent since 2015, our gross emissions have decreased by more than 60 percent during the same period.



* Net greenhouse gas emissions represents our total gross footprint minus carbon offsets. Percentages shown for each emissions category represent the share of Apple’s gross footprint. Annual avoided emissions may exceed the reduction in emissions from our baseline footprint due to various factors, such as business growth. Totals add up to more than 100 percent due to rounding.

** Low-carbon materials represents emissions savings from transitioning to recycled materials in our products and using low-carbon aluminum. For details, see [page 30](#).

† Clean energy represents emissions savings from clean energy procured by Apple or its suppliers.

* Business operations includes business travel, employee commutes, working from home, fuels and other energy-related activities, and the use of other cloud services.

Design and materials

Our purpose

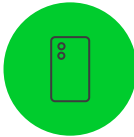
We design our products to be less carbon intensive by prioritizing the use of recycled and renewable content and low-carbon materials while focusing on the energy efficiency of our software and hardware. We’re working toward a future where every Apple product will be created from and contribute to circular supply chains. The design and material choices we make across our products also support reducing our carbon footprint.

Our path

Design and materials address emissions from:



Product manufacturing
(Scope 3)



Product use
(Scope 3)



Product transport
(Scope 3)

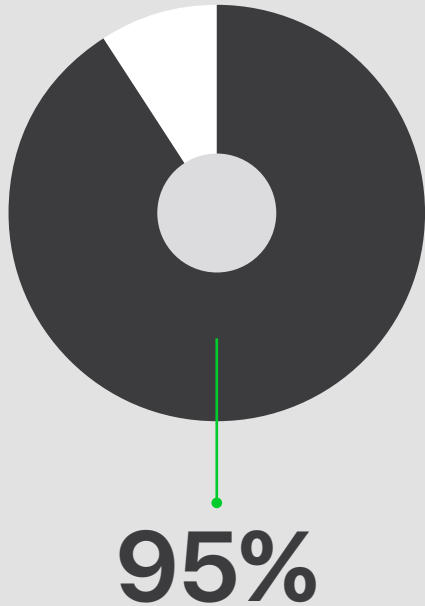
Our progress

More than doubled the use of certified recycled lithium and zinc shipped in our products in 2024²⁵



Our progress

iPhone 16 uses more than 95 percent recycled lithium in the battery, a first for Apple²⁶



Our progress

We now use 99 percent recycled rare earth elements in all magnets in our products



Our water strategy

Water is a local resource. For this reason, we have a context-based strategy that takes into account local conditions where we and our suppliers operate. We collect and analyze data and site-level feedback to understand our water impacts. We also gain insights into local watershed health — such as baseline water stress — using tools like the World Resources Institute (WRI) Aqueduct Water Risk Atlas.

We’ve identified that 70 percent of our corporate water use occurs in areas with high basin stress.⁵⁷ And we’ve found that the majority of our water-related impact is in the manufacturing supply chain.

We know that we can’t address water stewardship challenges alone. We believe we must go beyond our operations to collaborate with communities and work in basins on stewardship, replenishment, and WASH (water access, sanitation, and hygiene) projects throughout our value chain.

Low-water design

We approach low-water design by focusing first on site selection whenever possible and conducting a water risk evaluation to determine whether a potential site is in a water-stressed area. The results inform our decision-making and help mitigate the impact of our expected water use. We then design solutions to manage the quality of the wastewater that we return to the watershed.

Our low-water design efforts avoided 161 million gallons of freshwater use in 2024 in our corporate offices and data centers.⁵⁹ At our campus in Austin, Texas, we aim to achieve net-zero water use by relying on an onsite wastewater reuse system, as well as condensate recovery and stormwater capture, limiting potable water use to potable purposes only. This effort will save up to 60 million gallons of freshwater use annually.⁶⁰ Using wastewater for cooling will also allow us to achieve 28 million kilowatt-hours in energy savings annually at the location.

Our three most recent data centers have prioritized water efficiency by using high-efficiency air-cooled chillers instead of water-cooled systems, resulting in zero water use for cooling. At our newest data center in Waukee, Iowa, we’re also pursuing a 221-acre wetland restoration project on the property. The initiative will restore the land to native prairie potholes and wetlands, providing stormwater capture, groundwater recharge, and water quality benefits, as well as creating a habitat for native species.

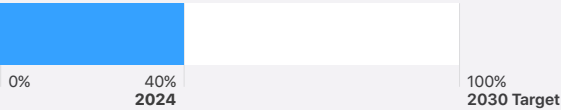
In our supply chain, the majority of water is used in manufacturing. We look for opportunities to innovate around water- and chemistry-intensive processes when possible. An example of this is a green anodizing project we initiated, which involved subject matter experts across internal design, product development, manufacturing, and quality teams, as well as an outside vendor piloting acid recovery equipment. The project integrated an acid recovery technology that separated and recycled two process chemistries — sulfuric and phosphoric acids — along with modular reverse osmosis systems that filtered water in the anodizing baths. The team worked to confirm that the approach functioned within both the tight process tolerances and the delicate chemistry required to maintain product quality. From 2020 to 2022, the collaboration facilitated this technology transfer — from early offline experiments and a proof of concept pilot to full production of a MacBook product.

The impact was profound, resulting in a 75 percent reduction in water usage and approximately 90 percent reduction in chemical use, which also helped achieve carbon savings.

Goal

Replenish all our corporate freshwater withdrawals in high-stress locations by 2030.

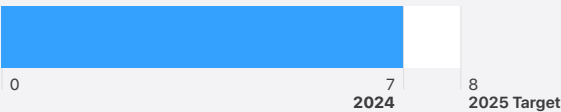
Progress



Goal

Certify all Apple-owned data centers to the Alliance for Water Stewardship (AWS) Standard by 2025.

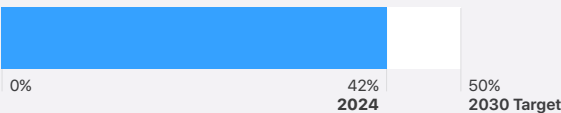
Progress



Goal

Increase supplier participation in the Supplier Clean Water Program, prioritizing locations with high water stress and supporting participants in achieving a 50 percent average water reuse rate by 2030.⁵⁶

Progress



We focus our work on five strategic pillars:



Low-water design

Minimizing water impacts in the design of products, Apple-managed services, and sites⁵⁸



Site efficiency and conservation

Improving the performance of existing sites and processes



Site water stewardship

Demonstrating responsibility beyond our facilities through watershed-level management



Replenishment and nature-based solutions

Improving water availability, quality, and access through regenerative approaches



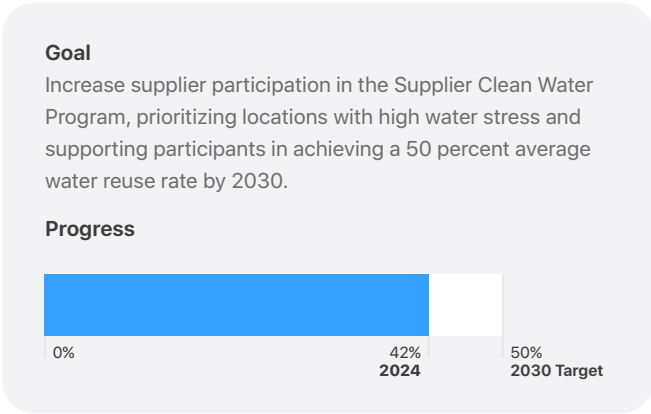
Leadership and advocacy

Advancing water management through policy, advocacy, and technology innovation

To learn more, read our white paper, [Apple’s Water Strategy](#).

Site efficiency and conservation

Next, we identify steps to use less freshwater in our existing operations and those of our suppliers, prioritizing regions where our efforts can immediately reduce stress on local watersheds.



In 2024, our facilities used about 1.8 billion gallons of water in our direct operations — a 9 percent increase from 2023. This was driven by growth in our corporate campus portfolio, as well as hot weather increasing irrigation and cooling needs. But through our site efficiency and conservation efforts, we saved 96 million gallons of freshwater.⁶¹ Last year, alternative water sources — primarily municipal recycled water — accounted for about 13 percent of our total corporate water usage. Additional efficiency and conservation accomplishments include:

- In early 2025, we completed cooling system controls optimization at our Maiden, North Carolina, data center, which is expected to save 7 million gallons of water per year. A second phase of this project is in progress.
- At our Parmer Lane campus in Austin, we performed retrocommissioning work that reduced the load on the cooling plant, saving over 3 million gallons of water per year.

At our data centers, server upgrades in 2024 saved 11 million gallons of water. We’ve also piloted a resin water treatment system that reduced makeup water use by 30 percent and discharge by up to 60 percent.⁶² Based on this success, we’re implementing this technology at our data centers in Prineville, Oregon, and Mesa, Arizona.

We also continued to use plant-based water treatment at several data centers, helping eliminate the need for biocides and corrosion inhibitors. This system, which uses compostable, natural sphagnum moss to improve water quality, can further support water savings. We implemented this at our Reno, Nevada, and Maiden, North Carolina, data centers and have begun a permanent installation in Mesa, Arizona.

Our supply chain accounts for 99 percent of our total water footprint, based on our detailed water inventory. We require our suppliers to maintain the high standards for water discharge outlined in the [Apple Supplier Code of Conduct](#). Through our Supplier Clean Water Program, we help suppliers minimize process water impacts and adopt best practices in water management and wastewater treatment. The program supports our suppliers in going above and beyond requirements to become stewards of the water resources where they operate by conserving water, promoting water reuse, and preventing water pollution within our supply chain. Since the program’s launch in 2013, the average reuse rate of more than 250 participating suppliers has increased to 42 percent, saving 14 billion gallons of freshwater in 2024 and over 90 billion gallons of water.⁶³ These savings come from a range of initiatives, such as the reuse of reclaimed water, upgrades to water-efficient equipment, and countercurrent rinse methods.

Site-level water stewardship

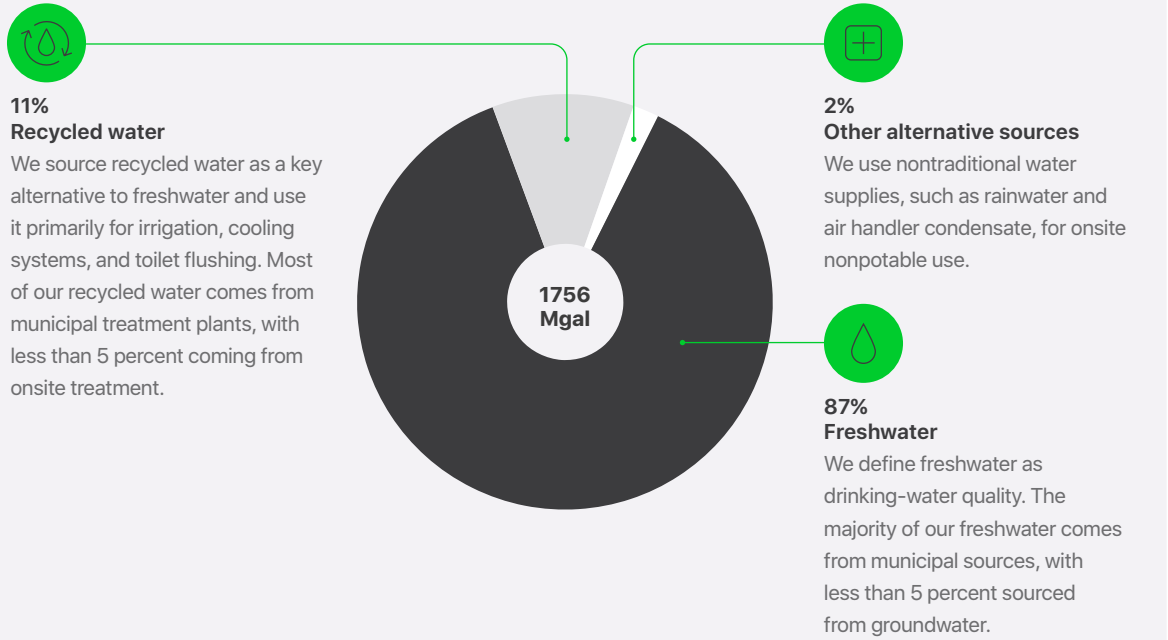
We engage with the communities around our facilities for the protection of local watersheds. Since 2018, we’ve partnered with the Alliance for Water Stewardship (AWS) to advance the AWS Standard, the first global framework to measure responsible water stewardship across social, cultural, environmental, and economic criteria. In 2020, we joined the AWS board of trustees to highlight water stewardship opportunities to our suppliers and promote collective action on shared water challenges impacting the sector.

Through AWS, we’ve certified seven of the eight data centers that we own and operate in Prineville, Oregon; Reno, Nevada; Maiden, North Carolina; Mesa, Arizona; Viborg, Denmark; and Ulanqab and Gui’an in China. We’re on track to certify our eighth data center in 2025.⁶⁴

Since 2018, 29 of our supplier sites have been certified to the AWS Standard, 24 of which have achieved a Platinum rating — the highest score within the AWS framework. Our work with AWS has allowed us to engage with suppliers at the regional level, focusing on the stewardship of basins with a concentration of manufacturing partners. In the Taihu Basin, proximate to Shanghai,

we’ve supported our efforts in the Supplier Clean Water Program and AWS certification by working with Water Champions. This program facilitates supplier field trips to water-sensitive areas within the basin. On these trips, participants experience first-hand lessons in water and wetland replenishment, as well as nature-based solutions for clean water flowing through the basin. These experiences reinforce the importance of efforts at the supplier sites by connecting individuals with the natural environment. In India, we also work with AWS and Frank Water to identify stewardship opportunities with our suppliers in the Bengaluru and Chennai basins, assessing the local water infrastructure and community needs.

Water use at corporate facilities



Replenishment and nature-based solutions

Replenishment and nature-based solutions are essential tools for addressing water availability, quality, and equity. Our goal is to replenish 100 percent of our corporate freshwater withdrawals in high-stress locations by 2030.⁶⁵ By the end of fiscal year 2024, we had contracts in place to address over 40 percent of what we expect to replenish.

2024 projects

Water Quality Protection and WASH in Northern Nevada

In the Truckee River Basin, home to our Reno Data Center, we partner with the local water utility, Truckee Meadows Water Authority. The project addresses water quality in the river and provides sanitation and water for drinking and hygiene to the downtown river corridor. The installation of a new Portland Loo in Reno City Plaza, along with continued operation and maintenance of freestanding public restrooms along the river in area parks, will prevent bacterial and nutrient pollution while providing access to dignified toilet facilities for the estimated 2000 people experiencing homelessness in Reno.

Progress: Apple’s support provided estimated water benefits of 70 million gallons in 2024 and is expected to result in a total of 468 million gallons through 2030.

Environmental Flows Restoration in Arizona

Our second project in the Colorado River Basin, home to our Mesa Data Center, supports ongoing efforts by The Nature Conservancy and its local irrigation partners to restore streamflow to the Verde River. The Verde Ditch canal piping project is expected to result in total of 3 billion gallons of water benefits, starting in 2025.

Groundwater Recharge and Bird Habitat in Northern California

In the Central Valley, we engaged with The Nature Conservancy on a second project, facilitating late irrigation season watering to recharge groundwater and support local bird habitat. Our collaboration with a Yolo County landowner and the Dunnigan Water District provided the majority of the water benefits in 2024. In Colusa County, the conservancy engaged with a commercial tomato grower to undertake field trials that added water to three different post-harvest tomato treatment methods. The conservancy and the growers aim to determine which method optimizes bird habitat and soil quality while accounting for water use, treatment costs, and impacts on the following year’s crop. The trial is aimed at determining the viability of

using drip-irrigated, harvested tomato fields for late-season bird habitat, which would provide the conservancy with access to an additional 200,000 acres of land to support threatened and endangered shorebirds that travel the Pacific Flyway through the Central Valley.

Progress: Apple’s support provided an estimated 58 million gallons of water benefits in 2024.

Wetland Restoration in the Chesapeake Bay Watershed

We partner with Ducks Unlimited to improve water quality in the Chesapeake Bay Watershed by reducing excess nutrient loading. Beginning in 2025, Apple will fund the restoration of degraded agricultural land to wetlands to improve water quality and support wildlife in the Chesapeake.

Progress: Our first project with Ducks Unlimited will reduce nitrogen, phosphorus, and sediment entering the Chesapeake Bay, resulting in total water benefits of 266 million gallons.

Village Water Supply in Tanzania

To support our replenishment needs in Africa, we’ve partnered with Water for Good to provide a safe, affordable, and lasting water supply to Bugogo, a village located in the Shinyanga Region of Tanzania. Water for Good is systematically addressing the region’s water supply, sanitation, and hygiene needs in partnership with the Tanzanian government’s Rural Water Supply and Sanitation Agency. In 2024, our funds supported the construction and maintenance of a water system that includes a solar-powered pump, which raises well water to a tank above the village before distributing it to 12 subvillage tapstands and taps at the village primary school and dispensary.

Progress: Water benefits are anticipated to grow over time, resulting in 21 to 42 million gallons in total water benefits.

Ongoing projects

Invasive Species Removal in Southern California

We expanded our efforts with the Council for Watershed Health to remove the invasive *Arundo donax* cane species in Southern California, home to our Culver City campus. Our support provided an estimated 13 million gallons of water benefits in 2024 and is expected to result in a total of 350 million gallons over the lifetime of the agreements.

Drinking Water Supply in India

We provided 40 million gallons of drinking water in partnership with the Uptime Catalyst Facility in India over the past two years. This work helped us meet our 100 percent water replenishment target for our corporate operations in India in 2023.

Forest Restoration in Arizona

We monitored ongoing restoration work under our 20-year agreement with the Salt River Project, where the Arizona Department of Forestry and Fire Management has more than 12,000 acres of the total 30,000 acres contracted or undergoing restoration. In 2024, the project provided water benefits of 4 million gallons and is on track to meet the expected total of 1.8 billion gallons of water benefits over the next 20 years.

Floodplain Restoration in Northern California

We continued our engagement with River Partners — a leader in large-scale, multi-benefit river restoration across California — as they prepare the conceptual restoration design for the nearly 1600-acre Dos Rios Norte property at the confluence of the Sacramento and Feather Rivers. In 2024, River Partners initiated a partnership with a local Native American Tribe that identified Dos Rios Norte as a site of ancestral significance. River Partners is collecting native seeds and propagating plants with traditional uses. Water benefits totaling 4.9 billion gallons are expected to begin in 2028.

Data

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 - High-quality carbon credits
 - Carbon footprint by product
 - Energy
 - Resources
 - Normalizing factors



Data

Greenhouse gas emissions

We account for our carbon footprint by following internationally recognized standards, like the World Resources Institute (WRI) Greenhouse Gas (GHG) Protocol and ISO 14040/14044.¹ Improving the accuracy of our carbon footprint is an ongoing process — as we learn more, we refine our carbon models and adjust our climate roadmap. We also regularly revisit the boundary of our carbon footprint as our data sources improve and our business evolves.

		Fiscal year				
		2024	2023	2022	2021	2020
Corporate emissions (metric tons CO ₂ e) ²	Gross emissions	666,800	471,400	324,000	166,380	334,430
	Scope 1	55,200	55,200	55,200	55,200	47,430
	Natural gas, diesel, propane	37,400	35,300	39,700	40,070	39,340
	Fleet vehicles	15,400	17,000	12,600	12,090	4,270
	Other emissions ³	2,400	2,900	2,900	3,040	3,830
	Scope 2 (market-based)	3,300	3,400	3,000	2,780	0
	Electricity	0	0	0	0	0
	Steam, heating, and cooling ⁴	3,300	3,400	3,000	2,780	0
	Scope 3	608,300	412,800	265,800	108,400	287,000
	Business travel	284,500	225,700	113,500	22,850	153,000
	Employee commute ⁵	152,700	164,100	134,200	85,570	134,000
	Other fuel and energy-related activities ⁶	166,400	18,300	10,600	0	0
	Work from home (market-based)	4,700	4,700	7,500	0	0
	Transmission and distribution loss (market-based)	0	0	0	N/A	N/A
	Other cloud (market-based)	0	0	0	0	0
Product life cycle emissions (metric tons CO ₂ e) ¹²	Carbon credits					
	Corporate carbon credits	-666,800 ⁷	-471,400 ⁸	-324,100 ⁹	-167,000 ¹⁰	-70,000 ¹¹
	Gross emissions (Scope 3)	14,500,000	15,570,000	20,280,000	23,020,000	22,260,000
	Manufacturing (purchased goods and services)	8,200,000	9,400,000	13,400,000	16,200,000	16,100,000
	Product transportation (upstream and downstream)	1,950,000	1,500,000	1,900,000	1,750,000	1,800,000
	Product use (use of sold products)	4,400,000	4,600,000	4,900,000	4,990,000	4,300,000
	End-of-life processing	70,000	70,000	80,000	80,000	60,000
	Carbon credits					
	Product carbon credits	-70,300	-13,500	0	-500,000 ¹³	0
	Total gross scope 3 emissions (corporate and product) (metric tons CO ₂ e)	15,110,000	15,982,800	20,545,800	23,130,000	22,550,000
Total gross carbon footprint (without offsets) (metric tons CO ₂ e) ¹⁴		15,300,000	16,100,000	20,600,000	23,200,000	22,600,000
Total net carbon footprint (after applying offsets) (metric tons CO ₂ e) ¹⁴		14,500,000	15,600,000	20,300,000	22,530,000	22,530,000

- Notes:
- For data on years prior to 2020, please reference past [Environmental Progress Reports](#).
 - Totals might not add up due to rounding.
- Apple’s carbon footprint boundary is aligned with the Greenhouse Gas (GHG) Protocol framework and includes emissions that are material and relevant to Apple, where data is available. Apple’s carbon footprint includes direct scope 1 emissions; indirect scope 2 emissions from purchased electricity, steam, heating, and cooling; and indirect scope 3 emissions from purchased goods and services, other fuel and energy related activities, transportation and distribution, business travel, employee commute, product use, and end of life.
 - Apple is carbon neutral for corporate emissions as of April 2020. Beginning in fiscal year 2022, we’ve expanded our footprint boundary to include scope 3 emissions associated with work from home, other cloud services, electricity transmission and distribution losses, and other fuel and energy-related activities.
 - Emissions from R&D processes.
 - Beginning in fiscal year 2021, we’re accounting for scope 2 emissions from the purchase of district heating, chilled water, and steam.
 - Beginning in fiscal year 2020, we updated our methodology for calculating emissions from employee commute to reflect employees working from home during COVID-19.
 - Impacts such as upstream emissions for scope 1 fuels and life cycle emissions associated with renewable electricity are included.
 - For a detailed breakdown of carbon offset purchases applied to our corporate footprint, see the carbon offsets table on the following page.
 - We retired 471,400 metric tons of carbon credits from the [Chyulu Hills project](#) in Kenya and [Guinan project](#) in the Guizhou Province of China to maintain carbon neutrality for our corporate emissions in fiscal year 2023. These projects are certified to the VCS and CCB standards.
 - We retired 324,100 metric tons of carbon credits from the [Alto Mayo project](#) in Peru and [Chyulu Hills project](#) in Kenya to maintain carbon neutrality for our corporate emissions in fiscal year 2022. These projects are certified to the VCS and CCB standards.
 - We retired 167,000 metric tons of carbon credits from the [Chyulu Hills project](#) in Kenya to maintain carbon neutrality for our corporate emissions in fiscal year 2021. This project is certified to the VCS and CCB standards.
 - We retired 70,000 metrics tons of carbon credits — 53,000 from the [Chyulu Hills project](#) in Kenya and 17,000 from the [Cispatá Mangrove project](#) in Colombia.
 - Because we’re committed to accuracy and transparency, we regularly refine our product life cycle assessment model and sources of data.
 - For fiscal year 2021, we retired credits from the [Chyulu Hills project](#) in Kenya and purchased carbon credits from two additional projects to offset a total of 500,000 metric tons of direct emissions across our value chain. The [first project](#), a REDD+ coastal conservation project in Guatemala, protects and conserves forests from deforestation and degradation. The [second project](#) aims to establish forests on about 46,000 hectares of barren land that isn’t otherwise in use across seven counties in the Guizhou province of China. Both projects are certified to the same high standards that we require for projects in the Restore Fund, including VCS and CCB standards.
 - Due to rounding, our gross and net carbon footprints do not always equal the sum of the subtotals disclosed above.

Data

High-quality carbon credits

We retired the following high-quality carbon credits toward our corporate and product emissions footprint for 2024.

Project name	Project description	Vintage	Volume retired (metric tons CO ₂ e)	Registry link
AF Forestal Apepu Expansion	Forestal Apepu S.A. is a company established in 2019 by an international forestry fund to conduct sustainable reforestation in Eastern Paraguay. The aim of the company is the sequestration of carbon and the production of quality timber in a highly deforested landscape. Forestal Apepu purchased two contiguous properties of 2658 ha in the Department of San Pedro. As most private properties in the region, the land was deforested decades ago and then used for agriculture and beef production. Through fast growing eucalypt plantations, trials of plantations with native species, and the strict protection of the remaining natural forest, Forestal Apepu aims at restoring forest cover. The company may expand even further in the future, upon identification of potential expansion areas in the region.	2021	73,093*	https://registry.terra.org/app/projectDetail/VCS/2369
Arbaro Forestal San Pedro	The VCS grouped project “Afforestation in cooperation with local landowners for Forestal San Pedro S.A.” represents one of the first major afforestation projects of the private sector in Paraguay. Forestal San Pedro is a company established in 2019 by an international forestry Fund. The company aims to establish and manage an area of 8000 ha of sustainable forest plantations for the sequestration of carbon and the production of quality timber in Eastern Paraguay. Forestal San Pedro seeks to cooperate with local medium and large landowners to conduct tree planting on their properties. As a diversification strategy, plantations are largely established in silvopastoral systems on traditional cattle farms. Landowners lease their land to the company and get a share of the expected benefits. At the same time, they continue using the land for cattle grazing in a manner compatible with forestry. The expansion strategy provides an entry point for tree planting in the degraded and deforested landscape of Eastern Paraguay, dominated by cattle grazing and mechanized soy production. Despite the need to develop new models that can address environmental degradation and climate change, while at the same time offering a sustainable supply to the national wood and timber market, the tree component is still far from integrated into the regional productive landscape. Target production areas encompass areas deforested decades ago that have been used for implanted pastures and mechanized soy, but are now degraded.	2019, 2020, 2021, 2022	78,507*	https://registry.terra.org/app/projectDetail/VCS/2361
Chyulu Hills	The Chyulu Hills REDD+ Project (CHRP) is a multi-partner initiative designed to promote climate change mitigation and adaptation, restore biodiversity, and create alternative livelihoods under the UN scheme of Reducing Emissions from Deforestation and forest Degradation (REDD+). It’s located in the Tsavo-Amboseli ecosystem in southeastern Kenya and stretches over an area of over 410,000 hectares. Its main geographic feature is the volcanic Chyulu Hills mountain range, from which the project derives its name. This project presents a broad ecosystem approach, including REDD+, to provide long-term sustainable financing and management to maintain the ecological integrity of an iconic African landscape. The project will help protect a very high-value wildlife and biodiversity area while supporting the development needs of Indigenous and other local communities.	2020	385,000	https://registry.terra.org/app/projectDetail/VCS/1408
Guinan	The Guinan Afforestation Project is located in the Guizhou Province of China and contributes to carbon removal and local sustainable development by planting trees on the barren lands. The project is planting across 46,000 ha on barren hills and degraded lands. The project activity aims to enhance biodiversity conservation by increasing the connectivity of forests, improve soil and water conservation, and generate income and job opportunities for local communities.	2019	100,500	https://registry.terra.org/app/projectDetail/VCS/2070
Guyana REDD+	Guyana’s Jurisdictional Forest Carbon Credit Program generates REDD+ carbon credits at a nationwide scale within the Amazon basin of South America, a critical watershed and hotspot of biodiversity. ART has issued 33.47 million TREES credits to Guyana for the five-year period from 2016 to 2020, which are available to buyers on the global carbon market for use toward voluntary corporate climate commitments. The jurisdictional program includes all 18 million hectares of forest in Guyana — about 85 percent of the landmass — and enables the country to benefit from its historically low deforestation rate, while funding low-carbon development priorities.	2019	100,000	https://art.apx.com/mymodule/reg/accview.asp?id1=1017

* This figure accounts for carbon credits applied to both product and corporate emissions for 2024.

Data

Carbon footprint by product

The following tables list the carbon footprints (in kilograms) of Apple products sold as of March 9, 2025, along with select configurations.¹

iPhone	Unit	Storage configurations			
		128GB	256GB	512GB	1TB
iPhone 16	kg CO ₂ e	56	61	74	–
iPhone 16 Plus	kg CO ₂ e	60	64	77	–
iPhone 16 Pro	kg CO ₂ e	66	72	84	95
iPhone 16 Pro Max	kg CO ₂ e	–	74	86	97
iPhone 16e	kg CO ₂ e	48	52	62	–
iPhone 15	kg CO ₂ e	56	61	74	–
iPhone 15 Plus	kg CO ₂ e	61	66	79	–

iPad	Unit	Storage configurations			
		128GB	256GB	512GB	1TB
iPad Pro 13-inch (M4) Wi-Fi + Cellular	kg CO ₂ e	–	107	120	–
iPad Pro 11-inch (M4) Wi-Fi + Cellular	kg CO ₂ e	–	93	107	–
iPad Air 13-inch (M3) Wi-Fi + Cellular	kg CO ₂ e	89	95	108	114
iPad Air 11-inch (M3) Wi-Fi + Cellular	kg CO ₂ e	76	83	96	–
iPad (A16) Wi-Fi + Cellular	kg CO ₂ e	74	77	86	–
iPad mini (A17 Pro) Wi-Fi + Cellular	kg CO ₂ e	65	71	–	–

Apple Watch ²	Unit	Select product configurations						
		Aluminum case with Sport Loop	Titanium case with Sport Loop	Aluminum case with Milanese Loop	Titanium case with Milanese Loop	Titanium case with Alpine Loop	Titanium case with Trail Loop	Titanium case with Titanium Milanese Loop
Apple Watch Ultra 2, carbon neutral	kg CO ₂ e	–	–	–	–	(12)	(11)	(11)
Apple Watch Series 10, carbon neutral	kg CO ₂ e	(8)	(8)	(8)	(8)	–	–	–
Apple Watch SE, carbon neutral	kg CO ₂ e	(7)	–	(8)	–	–	–	–

Note: Dashes indicate that the configuration does not exist.

1 Product carbon footprint data for Apple products are published in our Product Environmental Reports and are accurate as of product launch. In instances where carbon models were developed prior to product launch, we use preproduction units.

2 Greenhouse gas emissions prior to applied high-quality carbon credits are represented in parenthesis.